

PEC CH.1 Assignment C Basics of Communication System

Q.1 Write Electromagnetic (EM) wave spectrum, Frequency bands and their application.

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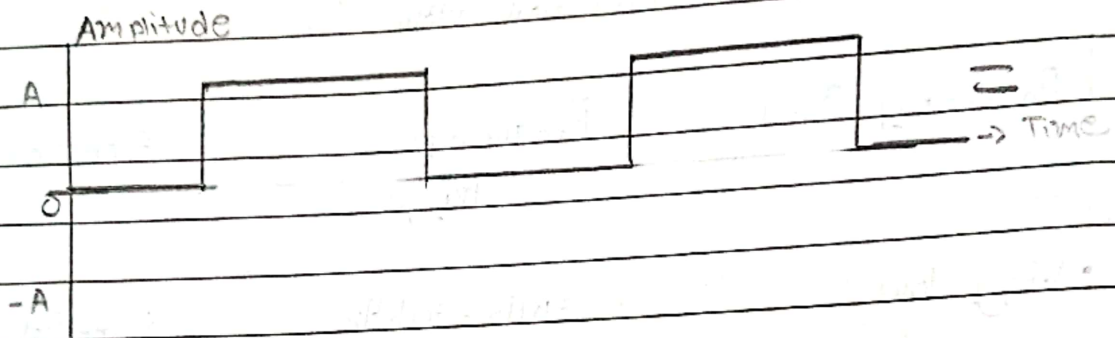
Electromagnetic Spectrum એ Wavelength અને Frequency બંનેના સંબંધમાં વર્ગીકૃત કરવામાં આવેલ છે. એલેક્ટ્રોમેગ્નેટિક રેડિયેશનની શ્રેણી છે.

Frequency Band	Frequency Range	Applications
• Very low Frequency	3 kHz - 30 kHz	Submarine communication, Navigation
• LF	30 - 300 kHz	Navigation, RFID, Amateur Radio
• MF	300 - 3000 kHz	AM Radio broadcasting
• HF	3 - 30 MHz	Shortwave broadcasting, TV broadcasting
• VHF	30 - 300 MHz	TV Broadcasting, Radio modems
• UHF	300 - 3000 MHz	Wi-Fi, Radar, TV broadcasting
• SHF	3 - 30 GHz	Radar systems, Satellite Commu.

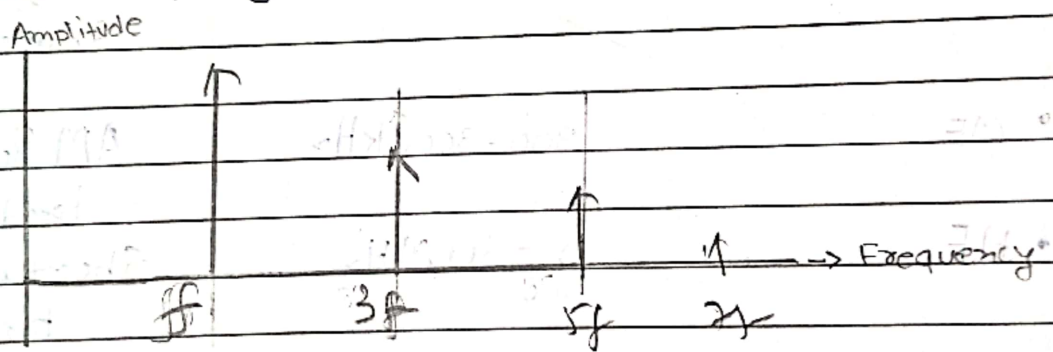
2. Draw Following Waveform in Time Domain and Frequency Domains

→ (i) Square Wave:-

Time Domain:- Signal fixed amplitude is HIGH and LOW level.

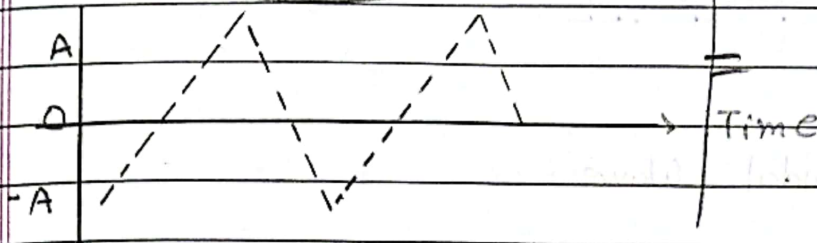


Frequency Domain:- Square wave has Fundamental Frequency and odd harmonics.



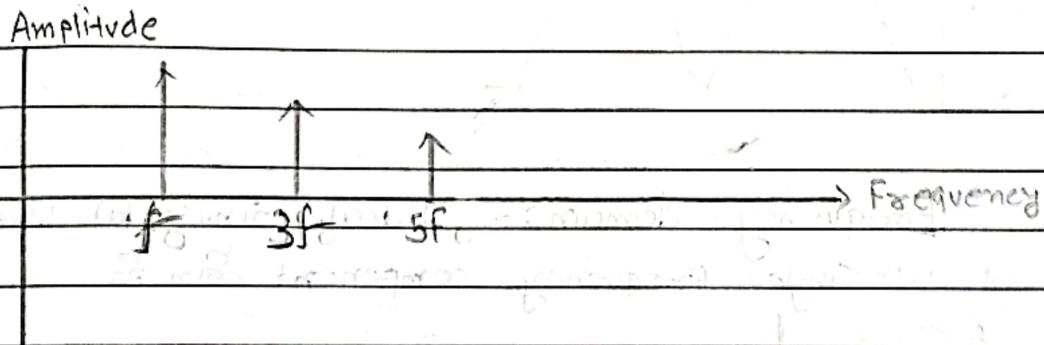
ii) Triangular Wave

Time domain:-



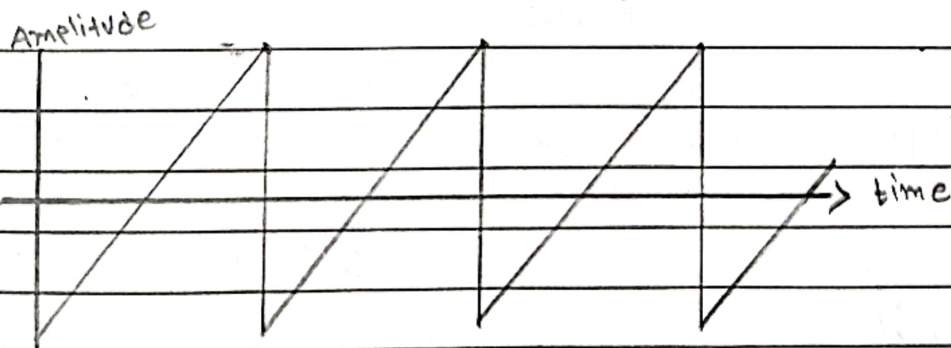
f = 0

Frequency domain:- Triangular wave has fundamental and odd harmonics. The higher harmonics have amplitude that decreases as frequency increases.

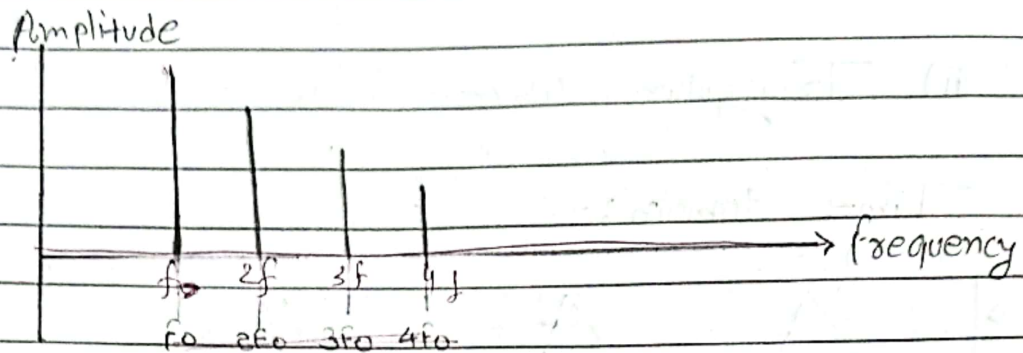


iii) Sawtooth Wave

Time domain:-

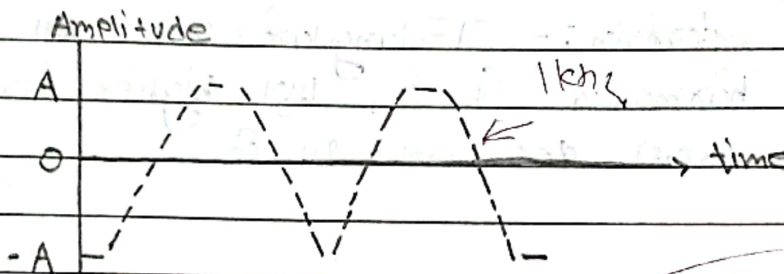


Frequency domain:- Sawtooth wave has fundamental and all harmonics present. The amplitude of each harmonic decreases as frequency increases.

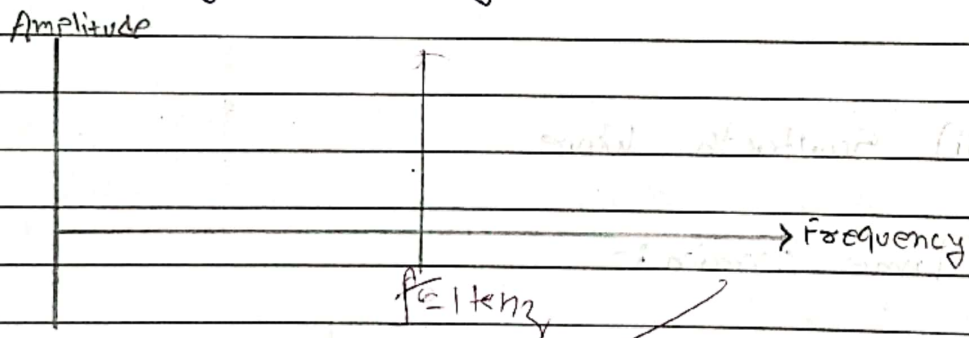


iv) Sinusoidal Wave

Time domain:-



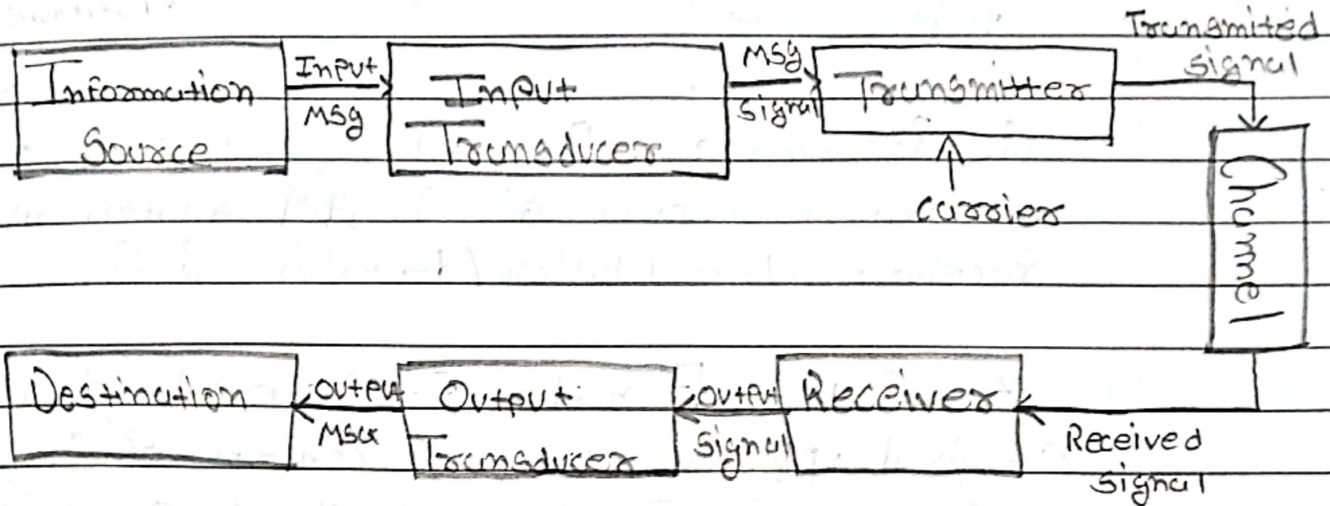
Frequency domain:- Ideal Sinusoidal wave is a single frequency component at f_0 .



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3. Draw and explain Block diagram of Analog Communication.

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• Explanation :-

1. **Information Source** :- Information transmit karani ke liye generate ki.

Ex. :- Human voice, music, video, data.

2. **Input Transducer** :- Physical information ko electrical signal me convert ki.

Ex. :- Microphone voice ko electrical signal me convert ki.

3. **Transmitter** :- Electrical signal ko transmission ke liye suitable karke aur aur signal amplification, modulation, filtering aur processes karke bhejti hai.

4. **Communication Channel** :- Transmitter aur receiver ke signal travel karne ka medium hai.

Ex. :- Air, Optical fiber, Coaxial cable.

5. Noise :- Channel में Unwanted disturbance signal में add हो जायेगा।

Ex. :- Electrical interference, Atmospheric noise etc.

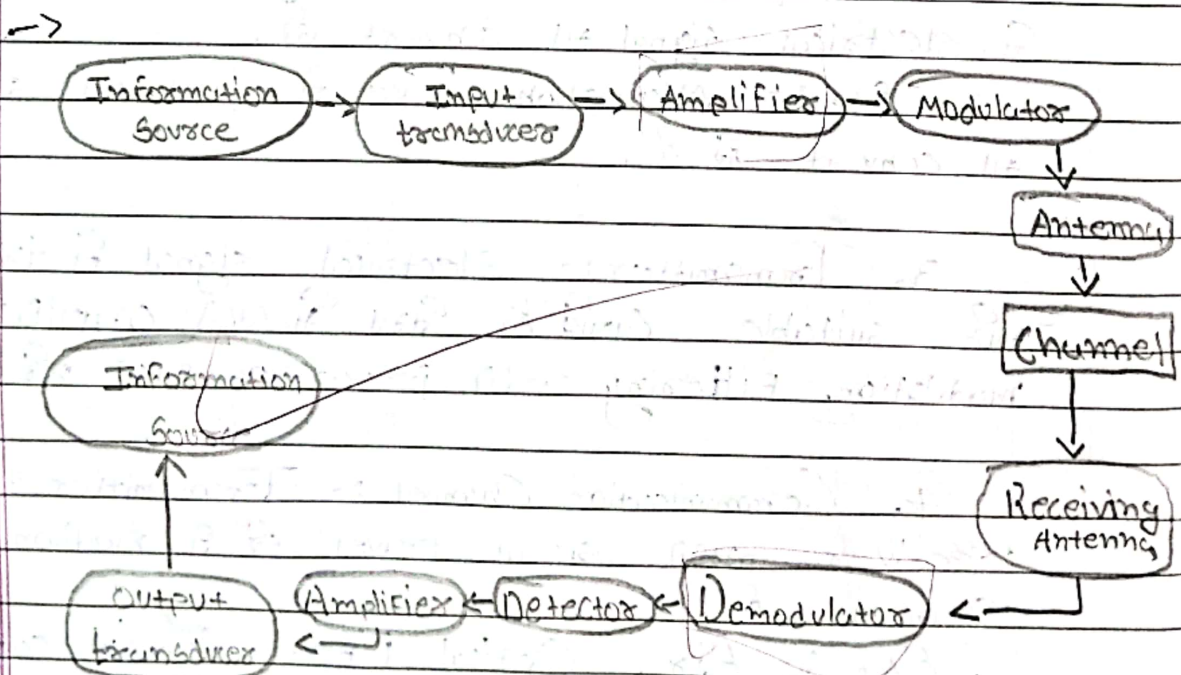
6. Receiver :- Received signal में original information recover की जायेगी। AM system में receiver demodulation/detection की जायेगी।

7. Output Transducer :- Electrical signal को original physical form में convert करेगा।

Example :- Speaker electrical signal को sound में convert करेगा।

8. Destination :- Recovered information final user/device में पहुँचाया जायेगा।

4. Draw and Explain block diagram of Digital Communication.



Explanation :-

1. Information Source :- Voice, video, text, image or data & information transmit through wire & provide to the receiver.

2. Input Transducer :- Physical information & electrical signal & convert to digital.

Ex. :- Microphone.

3. Source Encoder :- Information & digital form & convert to digital & unnecessary information remove & data compression.

4. Channel Encoder :- Data & extra bits add to the data & transmission & error detect & correct.

5. Digital Modulator :- Digital data & transmission & suitable signal & convert to analog.

Ex. :- ASK, FSK, PSK.

6. Communication Channel :- Signal transmitter & receiver & travel through the channel.

Ex. :- Air, Cable, Optical Fiber.

7. Digital Demodulator :- Received modulated signal & original digital data recover.

8. Channel Decoder :- Channel encoder માટે add કરેલ redundant bits ની use કરી errors detect કરે છે.

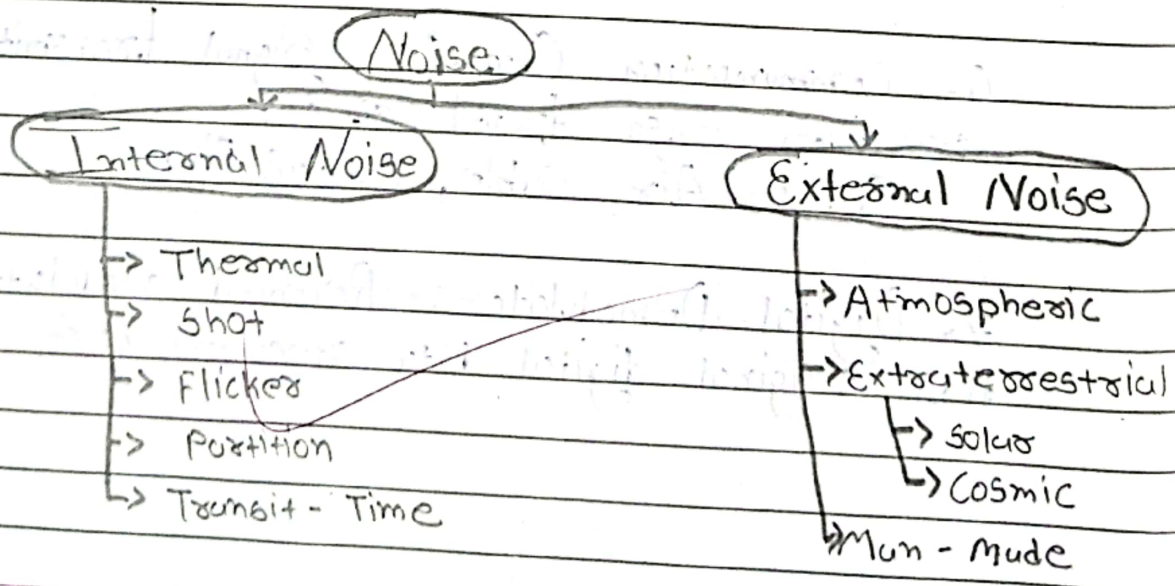
9. Source Decoder :- Compressed digital data ની decode કરી original information reconstruct કરે છે.

10. Output Transducer :- Electrical signal ની original Physical form માં convert કરે છે.
Ex. :- Speaker electrical signal ની Sound માં convert કરે છે.

11. Destination :- Final user/device જ્યાં recovered information પહોંચે છે.

5. What is Noise? Classify Noise and describe each one.

→ Noise ની Definition :- કોમ્યુનિકેશન System માં મૂળ સિગ્નલ સિગ્નલની પુર્ણ ભાવના કે કિસ્મીત કરવા કોઈ/વળા અવિચલિત ભાવના કેસ સિગ્નલની Noise કહે છે.



- Internal Noise :-
 - Thermal Noise :- તાપમાન ઉપરથી કારણે થતાં વિદ્યુતચુંબકીય ઇલેક્ટ્રોનિક્સના અનિવાર્ય પદોમાં random motion થાય છે, જેના કારણે આ noise generate થાય છે.
 - Shot Noise :- Transistor અને diodes જેવા active devices માં charge carriers continuous હોય છે પણ discrete રીતે flow થાય છે, જેના current માં random fluctuations થાય છે.
 - Flicker Noise :- Electronic Components માં low-frequency range માં થતાં મુખ્ય Noise છે.
 - Partition Noise :- જ્યારે circuit માં current ૨ કે તેથી વધુ અલગ-અલગ paths માં divide થાય છે, ત્યારે આ random current distribution થી noise create થાય છે.
 - Transit-Time Noise :- High frequency signals માં જ્યારે charge carriers ની junction cross કરવાનો time signal ના time period જેટલો થાય ત્યારે આ noise produce થાય છે.
- External Noise :- આ બહારના Communication system માં અવરોધક sources થી generate થાય છે.
 - Atmospheric Noise :- કુદરતી બળાવણો માં થતાં ઇલેક્ટ્રિકી જેવાં lightning અને storms ના કારણે generate થાય છે.
 - Extraterrestrial Noise :- બહારના બળાવણો ના કારણે થતાં noise.
 - Solar Noise :- Sun થી આવતાં radiations ના કારણે.

° Cosmic Noise :- Stars and galaxy and other celestial bodies emit various radiations called.

° Man-Made / Industrial Noise :- Motors (giriya) equipment and automobile ignition high voltage power lines and heavy industrial machinery are called ~~noise~~ ^{sources}.

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